# Valve Inventory

User Manual

A task-by-task walkthrough of the desktop application. For the command-line equivalent of anything here, see the Technical Manual's command reference, or README.md.

## Overview

The window opens on five tabs, described below, sharing one database. Nothing you do in one tab is hidden from the others - move stock in the Valves tab and the Browse tab's counts update the next time you search it.

- **Valves** - search, edit reference data, add/take/move/delete stock.
- **Bases / Sockets** - the same idea, for the sockets themselves rather than the valves that plug into them.
- **Browse** - a parametric filter across every held type, for “what do I have that could work here” questions.
- **Repair Bench** - for “I've got this valve out of a set I'm fixing - what is it, and what have I got that could stand in for it?”
- **Docs** - a general reference library for material that isn't about one specific type - care-and-feeding guides, base wiring references, and the like.

### English or Portuguese

Two flags sit at the top right of the window. Click either to put the whole interface into that language - menus, tab names, buttons, column headings, filter captions, dialogs, the About box and the user guide all change together. The choice is remembered for next time, and switching costs nothing: no window is rebuilt, so your filters, the selected box and any open popup stay exactly as they were.

What does *not* change is the collection. Type designations, box names, makers, origins and your own notes stay exactly as they were typed - a valve is an EL84 in any language, and a bag labelled “Saco Pingo Doce” is called that because that is what is written on it. The filter dropdowns are filled from the database, so their contents stay in the language of the data too. Only the tool's own wording moves.

## The Valves tab

### Boxes sidebar

Click a box to filter the results to it; click “All boxes” to clear. Click a column heading (Box / Types / Qty) to sort, click again to reverse.

### Search row

Text, Function, Base, and the numeric fields (Heater V, Pa W, Freq MHz), which accept comparisons like `>20`, `<7`, `>=250` as well as an exact value.

> **Note:** Searching a type by name also pulls in stock of anything cross-referenced as its equivalent - search ECF80 and PCF80 stock shows up too, in blue, labelled which type it's equivalent to. This is a hint toward substitutes, not a claim they're necessarily interchangeable in every circuit.

**Advanced...** opens a dialog covering every remaining field - manufacturer, condition, family, gm, mu, power out, confidence, and whether a datasheet is linked. It edits the same underlying filters as the quick row, so the two stay in sync.

## Results table

Click a heading to sort. Double-click a row to open its datasheet. **Amber** rows have unconfirmed (inferred) parameters; **blue** rows are equivalents pulled in by a search.

## Detail panel

The selected row's type record, editable in place. **Save** keeps it inferred; **Save + confirm** marks it confirmed and the row turns from amber to black - that transition is the progress bar for working through the collection.

Below the fields, **Similar types** lists other held types with the same broad function and every shared electrical rating within 50% - candidates for a substitute with modification, not verified equivalents. Heater mismatches are flagged, not filtered out, since a dropping resistor or a different supply can often cover that. Double-click a suggestion to look it up.

**Open datasheet** opens the local PDF if there is one, otherwise falls back to an online source - the button itself reads *Open datasheet (local)* or *Find datasheet (web)*, so which one it'll do is clear before you click. **RadioMuseum** and **Web search** run a site-scoped or general search for whatever's selected. **Manage...** (**Manage information...** on the Browse tab's popup) opens the full document list for a type: the one “primary” sheet that button opens, plus as many extra datasheets and links as you like - a second manufacturer's sheet, a forum thread, a project that happens to use this valve. Upload a file you already have, or paste a URL (no download needed for a link - it's just recorded). Its **Edit parameters...** button opens the same field-entry form as the detail panel, so a Browse-tab research session never needs to switch tabs to record what a datasheet says.

## Toolbar

**Add stock** creates the type automatically if it's new, classifying it from its designation. **Edit lot**, **Individual valves...**, **Take**, **Move**, and **Delete lot** act on the selected row. **Move** also offers a position in the destination box; leaving it blank clears the old one, which belonged to the box the lot has just left.

> **Note:** The two editors either side of the results table do different jobs. **Edit lot** changes this one physical lot - where it is, what it came from, how it tested. The panel on the right changes the reference record shared by *every* lot of that type.

# What a lot records

A lot is one physical batch: this many of this type, in this box. Two Mullard EL84s out of different sets are one *type* but two *lots*, and it's the lot that knows which shelf it sits on and which set it came out of. Beyond quantity, manufacturer and condition, each lot can record:

| Field | What goes in it |
| --- | --- |
| Position | Where in the box it sits, as a grid reference - B-12, row and column. |
| Type 1 / Type 2 | Other designations the valve is marked with: a US number, a service code, a second maker's part number. |
| Origin | Where it came from - bought, inherited, or the set it came out of. |
| Test values | What it measured on a tester. |
| Other | Anything else: boxed or unboxed, odd printing, whatever the row needs. |

**Every one of them is optional**, and blank is a perfectly normal value - none of them changes how anything else behaves. Fill them in from **Add stock**, from **Edit lot** afterwards, from the upload CSV, or from the command line with `valves.py add` / `valves.py edit`.

Type 1 and Type 2 sit on the lot rather than the type on purpose: they record what *this* glass is actually marked with, which is not always the designation you file it under. They're searchable either way, so a valve stored as EL84 and printed 6BQ5 is found by either name. That's separate from the type record's *equivalents* list, which is a claim about the types themselves rather than about one batch's printing.

Position is a plain grid reference rather than two separate row and column fields, so it fits whatever scheme a box already uses - B-12, 3/4, or a shelf name. Lot listings sort by it, with un-positioned lots last, so a partly-positioned box still reads top to bottom.

On the command line, a listing leaves out any of these columns that's empty for every row it's showing, so `valves.py box 12` looks exactly as it always did until there's something in there to show. In the window they're always-present columns on the results table, since the table scrolls sideways and a stable column layout is easier to work against.

# Individual valves and testing

A lot is a quantity - “6 x KT66 in box 8” - and for most of a collection that is all it ever needs to be. Where it isn't, select the lot and click **Individual valves...**, then **Track individually**. That creates one row per valve held, and from then on each valve is a thing in its own right: its own position on the shelf, its own serial or date code, its own maker and condition where a lot is mixed, and its own test history.

Expanding a lot is opt-in and per lot, so a box of a hundred identical indicators stays one line until you decide otherwise. It is also safe to repeat - it only ever tops a lot up to the quantity it holds, never duplicates or resets what is already there. The **Ind** column on the results table shows how many of each lot are tracked this way; blank means the lot is still just a quantity. New lots added with **Add stock** are tracked individually from the start unless the form says otherwise.

The **Notes** column in that list is what was written about that one valve - a serial read off the glass, “no box”, “another one at home”. It belongs to the valve rather than the lot, which is the whole point of tracking individually: a remark that applies to one valve out of six says nothing useful once it has been pooled onto all six.

## Recording a test

**Record test...** logs one test of the selected valve. Every reading is optional, because no single tester produces all of them: an emission tester gives one figure, an AVO VCM163 reads anode current and mutual conductance on two meters at once plus separate gas and insulation tests, a curve tracer gives everything. A record holding nothing but a gm figure and a date is a perfectly good record.

| Field | Units | What it is |
|---|---|---|
| Tested on, Tester | | When, and on what. A test dated 1901-01-01 is one recovered from a written record that gave no date - nothing was tested in 1901, the valve had not been invented, so the date reads unmistakably as "tested, date unknown" rather than as a real measurement day. |
| Va, Vg at test | V | The conditions the readings were taken under. A gm figure means nothing without them. |
| Bias mode | | Fixed or auto. The same valve reads differently under each. |
| Ia | mA | Anode (plate) current - the headline figure on most testers, and what power valves are matched on. |
| Ig2 | mA | Screen current, for tetrodes and pentodes. |
| gm | mA/V | Mutual conductance. British practice throughout; multiply by 1000 for the micromhos an American tester shows. |
| gm as % of nominal | % | How valves are actually graded and sold. |
| Emission | % | The single reading a cheap emission tester gives. |
| Gas / grid current | uA | The gas test - an AVO reads to 100 uA full scale. |
| Insulation | Mohm | Interelectrode leakage. |
| Heater-cathode | | The separate cathode/heater test: a figure, or pass/fail. |
| Shorts, Verdict | | Pass/fail, and your overall call on the valve. |

**Note:** Testing is never destructive. Each test is a new row, so retesting a valve years later builds its history rather than replacing it - and the trend between two readings is usually the interesting part. **Test history...**, or a double-click on the valve, shows every test of it, newest first.

The tester and the test conditions are carried forward from that valve's last test, since they rarely change across a session and the readings always do.

### *Double triodes*

A double triode is recorded a section at a time: run **Record test** twice, once with Section *a* and once with *b*. That is how the readings come off the meter, and comparing the two sections is the whole point of testing an ECC83 for phase-inverter duty. The valve list shows the most recent test of either section; the history shows both.

## Colours in the valve list

**Amber** rows are valves that have never been tested. **Red-brown** rows are ones whose last verdict was weak, short or failed.

## Using valves up, and correcting the record

These are two different things and they behave differently. **Take**, on the Valves tab, is for a valve you have actually used: it reduces the lot's quantity and removes that many individual rows as well, choosing the *least documented* first - untested before tested, unmarked before serial-numbered - so using valves up never quietly discards test history you took the trouble to record. **Remove valve**, in the dialog, is for a row that should not have been there: it deletes the record and leaves the quantity alone.

> **Note:** Deleting a valve takes its test history with it, and deleting a lot takes its valves and their tests. That is deliberate - a test belongs to a particular piece of glass and means nothing without it - but it is the one irreversible action here, so the dialog says what will go before it goes.

**Tools > Check individual valve counts** reports any lot where the quantity and the number of individual rows have drifted apart. It reports rather than corrects: which of the two is right depends on what is actually in the box.

# The Bases / Sockets tab

Valve bases and sockets aren't valves, so they're tracked in their own table rather than mixed into the general sundry catch-all. Same pattern as the Valves tab: search by base type or box, Add / Take / Move / Delete lot.

# The Browse tab

A faceted filter across all held types, closer to a shopping-site filter panel than a search box.

- **Category, Base, Family, Confidence, Variable-mu, Tested** - dropdowns that *cascade*: picking one narrows what the others still offer, so you never land on an empty combination. Category is a coarser bucket than the raw function text (Triode, Double triode, Tetrode, Pentode, Triode-pentode, Rectifier, and so on) - specifically so it's useful as a filter instead of nearly matching one type each.
- **Numeric ratings** (Heater V/A, Va max, Pa max, gm, mu, Power out, Freq max) - an operator (< = > <= >=) plus a value picked from what's actually present in the data.
- **Name contains** - narrows the list as you type, e.g. “3cx” or “PL”.

Click a heading to sort. **Double-click a type** for a popup showing its full reference record, datasheet/web-search buttons, and a box-by-box breakdown of exactly where and how many you hold.

In that popup, **double-click one of the box rows** to drop straight into the individual valves of that lot. It is the same window the Valves tab reaches through **Individual valves...**, so a valve found by browsing behaves exactly like one found by searching - you can read its notes, see its test history and record a new test without going back to the Valves tab.

The **Tested** facet narrows the list to types that hold at least one tested valve, or to those that hold none, and the **Tested** column counts them. The Valves tab carries the same filter for lots, beside its numeric fields, with a **Tstd** column next to **Ind**. Both count a lot or a type as tested when at least one valve in it has at least one recorded test; a lot with no individual valve rows at all therefore reads as untested, because nothing in it has been tested.

# The Repair Bench tab

The workflow for a valve pulled out of a set on the bench: type its designation (and, optionally, which circuit stage it came from - IF amp, audio output, rectifier, and so on), then **Identify**.

## If it's already in your database

Its reference data loads straight into the form on the left. On the right, **In stock now** shows anything you already hold of that exact type or a listed equivalent, and **Possible substitutes** lists other held types with the same broad function and every shared rating within 50% - the same candidate logic as the Valves tab's Similar types, but scoped to what's actually in stock, and with a held-quantity count. Double-click a substitute to switch the whole bench over to it, if that turns out to be the more interesting question.

## If it's new to you

**Open datasheet**, **RadioMuseum**, and **Web search** work immediately off the typed designation, before anything is saved. **Copy research prompt** puts a ready-to-paste prompt on the clipboard, scoped to just this one type (a faster, single-item cousin of Tools > Generate research prompt...) - paste it into Claude, and it comes back in the same block format Apply researched data... expects.

**Add to database** creates a bare reference record (classified from the designation, no stock attached) so there's somewhere to save findings as you gather them. **Save** / **Save + confirm** work exactly as in the Valves tab detail panel - and immediately refresh the substitute list on the right using whatever you just entered, so you can see straight away whether the parameters you found open up any new candidates from stock.

# The Docs tab

A general reference library, for material that isn't about one specific valve type - a care-and-feeding guide for power tubes, a base wiring reference, anything worth keeping alongside the collection. **Add from file...** copies a PDF you already have into the local archive; **Add from URL...** just records a link, no download. Each entry gets a title and an optional abstract - select one to read its abstract in the pane on the right, and use the filter box to narrow the list as you type.

> **Note:** The same title/abstract/file-or-URL idea also applies per type, via the Valves tab's or Repair Bench's **Manage...** button - the difference is only whether a document is filed against one valve type or kept in the general library.

# Filling in reference data

New types start out with only what the naming convention can infer - a real datasheet reading is what actually confirms them. Three ways to close that gap:

## By hand

Edit the fields in the detail panel and Save + confirm, as above.

### With Claude - electrical parameters

Tools > Generate research prompt... writes a prompt (for your highest-quantity unconfirmed types) to a text file. Paste it into any Claude chat, save the reply, then Tools > Apply researched data.... Only what Claude actually confirmed is applied - a hedged finding (“could not verify”, “plausible”) is kept as a lead rather than marked confirmed, so nothing gets overclaimed.

### With Claude - datasheet files

Tools > Generate datasheet download prompt... writes a prompt aimed at an agent with file and web access (Claude Code, not a plain chat, since it needs to write files to your disk). It tries the built-in fetcher first, then searches further for whatever's still missing, and saves PDFs directly into the local archive.

## Adding stock in bulk

For more than a few lots at once, skip the Add-stock dialog:

- **Tools > Create upload template...** writes a blank CSV with the right columns, ready to fill in.
- **Tools > Import upload CSV...** reads a filled-in CSV back in - one row per lot, new types classified automatically, existing types just get more stock.
- **Tools > Generate CSV-building prompt...** writes a prompt for any Claude chat that interviews you (or reads whatever spreadsheet, notes, or photos you describe) and hands back a ready-to-import CSV - useful when your existing records aren't already in this shape.

## Backup, export, and sharing

### Backup

The database itself isn't the backup - the text snapshot is. Refresh it before ending a session of changes:

```
python3 snapshot.py
```

That's what belongs in version control; it's what a restore rebuilds from.

### Export a spreadsheet

File > Export spreadsheet... writes a plain .xlsx for anyone who just wants to look, not use the tool.

### Hand the whole thing to someone else

File > Export archive and tools (.zip)... bundles the code, docs, and a fresh snapshot into one file, with an option to strip the third-party descriptive text first (see the Technical Manual's note on that). The recipient unzips it and follows QUICKSTART.md, which is included.